

- 6 Students are learning about a simple check digit method for data validation. In this method, a single check digit is appended to the end of an original number to give a new number.

The students are studying a method which:

- calculates the sum of all the digits in the original number
- uses integer division to calculate the remainder when the sum is divided by 10
- uses the remainder as the check digit
- appends the check digit to the original number, creating the new number.

For example:

original number	4162
sum of all digits	$4 + 1 + 6 + 2 = 13$
remainder when the sum is divided by 10 using integer division	3
new number	41623

The method described can be used to detect single-digit errors. For example, if the new number is incorrectly input as 41633, then the check digit does not match and the input will be rejected.

- (a) An incorrect attempt was made to enter 41623. The **first two** digits were entered incorrectly; the **last three** digits were entered correctly.

When this number was tested, the check digit was found to be correct and the input was accepted.

Identify an example of the incorrect attempt to enter 41623 and explain why this number would **not** be rejected.

Number

Explanation

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[2]

